

▣ 19 Sampling

Inferential statistics is concerned with taking a sample, observing a pattern or structure in the sample and generalising or inferring that the same pattern or structure will be evident in the population.

For inferences to be valid, an appropriate mechanism needs to be used to select the sample

▣ **Probability sampling v Non-probability sampling**

Probability sampling (sometimes called random sampling) is a sampling mechanism whereby every individual in the population has a known, non-zero probability of being selected in the sample.

Probability sampling should be used if we want to have confidence in the inferences drawn from the sample to the population.

Examples of probability sampling methods include simple random sampling (SRS), systematic sampling and stratified random Sampling.

Examples of non-probability sampling include convenience sampling and quota sampling

Probability sampling

1. SRS

SRS is simple in principle. For example, a SRS of 30 Irish companies from the population of all Irish companies could be obtained by obtaining a list of the names of all Irish companies (a sampling frame), placing all the names in to a hat and drawing 30 without replacement.

Often there may be considerable logistical problems with obtaining a SRS in practice.

Sampling frames may not exist, or you may not be permitted to access them.

Even if this problem is surmountable, there is often considerable effort (and expense!) required to locate the particular candidates selected for the sample.

2. Systematic sampling

A sample of 500 people may be selected using a telephone book by randomly selecting a starting entry and then selecting say every 200th entry thereafter.

Note that the population in this case is of those whose number is listed in the telephone book.

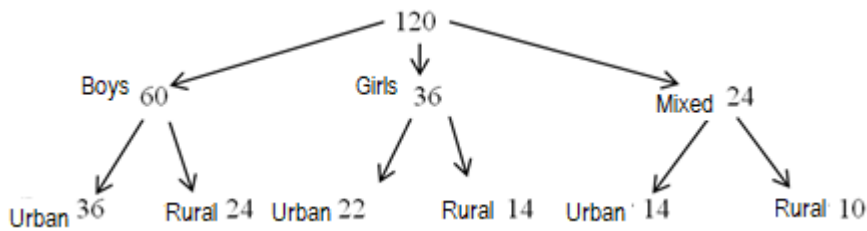
Systematic sampling is more convenient than SRS but if there are cyclic phenomena (e.g. in a production process a sample of output is required for quality testing and every 5th item is made by the same operator) then our sample will not be representative.

3. Stratified Random Sampling

Suppose a sample of 120 primary schools in a country are to be surveyed. The gender make-up in the population of such schools is one of all-male (50%), all female (30%) or mixed (20%). Additionally, 60% of schools are in urban areas and the rest in rural areas.

Stratified sampling would see the sample of 120 made up of 60 (50%) from boys schools, 36 (30%) from girls schools and the rest 24 (20%) from mixed schools. Within boys schools, the 60 would be made up of 36 (60%)

from urban areas and the rest 24 (40%) from rural areas. The rest of the sample would be constructed in a similar way.



Within each strata, a basic probability sampling method (e.g. SRS) to obtain the sub-sample.

While SRS would ensure that the various strata were represented proportionally on average, stratified sampling will do so in each and every sample.

4. Multistage Cluster Sampling

In household surveys, in particular, if interviewer-assisted surveys are considered necessary, then interviewer travel and subsistence costs can become a major issue. So SRS may not be practical and an alternative often used is multistage cluster sampling

A random sample of geographical clusters of households is randomly selected (stage 1), then a random sample of households within each of these clusters is selected (stage 2) and finally an individual within the household is randomly selected in the sample (stage 3).

□ Non-probability sampling

5. Convenience sampling

Convenience sampling essentially involves arbitrary selection. For example, to obtain a sample of output from a process a single batch of output might be obtained and the required number of units obtained from that batch.

6. Quota sampling

Quota samples are more usually employed in social science (surveys etc.) where the sample must meet certain quotas e.g. 40 males, 30 urban etc, but the quotas can be filled in the most convenient manner.

Hierarchical data and independence

□

Suppose a sample of national school children in Waterford county is required for some research end. The sample might be obtained using something similar to multi-stage cluster sampling as follows.

First, a number of towns/villages are selected (possibly using convenience rather than random sampling). Within each school, perhaps one class might be selected. All children in that class are used in the sample.

There clearly is a hierarchical / clustered structure to such a sample. Children in one class (and therefore, school and town) will be more similar to children in a different class, school and town.

In particular children in one class cannot be reasonably assumed to be independent. Having measured one student (e.g. say on an ambition scale), that conveys something about what other students in the class might measure.

N.B. An assumption of the basic inferential methods that we will consider are that the data are independent.

Thus, it would not be appropriate to use such tests on data collected in this way.

Random sampling v random assignment

Take care not to confuse the concepts of random sampling and random assignment. Above, we have discussed random sampling. Random assignment is a consideration in experimental studies where the investigator may allocate subjects to treatment groups.

For example, a study to explore the possible impact of students listening to music while studying on the performance in an exam might randomly assign one third of participating students to listen to classical music, one third to popular music and one third to no music at all (the control group).

Opting for random assignment rather than allowing the students to self-select will tend to reduce the risk of a confounding variable confusing any findings.

I.e. perhaps smarter students might be more inclined to listen to classical music and of course smarter students are certainly more likely to perform better in an exam. An investigator might then erroneously report that listening to classical music led to better exam performance.

Statistics = Probability ?

Statistics and probability are often presented together in one module as they are here. There is a tendency to regard statistics and probability as being the same thing as both involve the mathematics of uncertainty.

While the former makes much use of the latter they are in fact distinct disciplines in applied mathematics. Indeed, probability (or the relative frequency view of it) can be seen in a certain sense as the reverse procedure of a problem in inferential statistics.

To illustrate, imagine that we knew that say 30% of an adult population are smokers from which we selected 50 people at random and asked, how likely is it that at least 15 of the 50 are smokers. This is a problem in the realm of probability.

On the other hand, if we did not know the percentage of smokers in the population we might estimate this percentage by obtaining a sample of adults and inferring that the percentage of smokers in the population is the same as in the sample. This is a problem in (inferential) statistics.

Sample knowledge	→	(inferential) Statistics	→	Acquire new information on population
Population knowledge	→	(rel. freq.) Probability	→	Acquire information on a sample